

F1 PitWall AI: An Intelligent Race Strategy and Decision Support System

AI-based web application that helps users understand and analyze Formula One race strategies using historical race data. The system collects information such as lap times, tire compounds, pit stops, weather conditions, and driver performance from free and publicly available sources. It processes this data to provide useful insights, race analysis, and strategy recommendations through an easy-to-use dashboard.

The main goal of this project is to demonstrate how Artificial Intelligence and Machine Learning can be used to solve real-world problems in Formula One racing. The system allows users to compare race strategies, predict tire performance, recommend the best pit-stop timing, and explore different race scenarios. It is designed as a learning and analysis platform for students, researchers, developers, and Formula One fans, while showcasing practical skills in AI, data analytics, and software development.

Tech Stack:

Python, FastAPI, Streamlit, PostgreSQL, FastF1 API, Ergast API, Scikit-learn, XGBoost, Pandas, NumPy, Plotly, Matplotlib, Git, GitHub, Visual Studio Code, and Streamlit Community Cloud/Render for deployment.

Project Objectives

- To develop an AI-based system for analyzing Formula One race strategies.
- To collect and process historical Formula One race data from free public sources.
- To predict tire performance and recommend optimal pit-stop timing.
- To compare different race strategies using data-driven analysis.
- To provide an interactive dashboard for visualizing race statistics and insights.
- To simulate "what-if" race scenarios for better strategy evaluation.
- To demonstrate the practical application of Artificial Intelligence, Machine Learning, and Data Analytics in Formula One racing.

Dataset

The **F1 PitWall AI** project uses historical Formula One race data collected from free and publicly available sources. The primary datasets are obtained from the **FastF1** Python library and the **Ergast API**, which provide detailed race information for analysis and machine learning.

Dataset Includes:

- Driver information
- Constructor (Team) information
- Circuit details
- Race schedules and results
- Lap-by-lap timings
- Sector times
- Tire compounds
- Pit stop information

- Weather data (where available)
- Driver standings
- Constructor standings
- Qualifying results
- Sprint race results (where available)

Dataset Sources:

- **FastF1** - Provides lap times, telemetry, tire data, weather, and session data.
- **Ergast API** - Provides historical race results, driver details, constructor information, circuits, and championship standings.

The collected data is cleaned, processed, and stored in a database before being used for data analysis, machine learning, and race strategy recommendations.

Project Implementation Information

Module 1: Data Collection

- Collect race data using FastF1.
- Fetch historical race results using Ergast API.
- Store raw data in a database.

Output:

- Clean historical race dataset.

Module 2: Data Preprocessing

- Remove missing values.
- Convert data into a standard format.
- Extract useful features.
- Prepare datasets for machine learning.

Output:

- Processed dataset ready for analysis.

Module 3: Database Management

Store:

- Drivers
- Teams
- Circuits
- Races
- Lap times

- Tire compounds
- Pit stops
- Predictions

Output:

- Structured database.

Module 4: Race Analytics

Display:

- Driver statistics
- Team performance
- Lap comparisons
- Sector analysis
- Tire usage
- Pit-stop history

Output:

- Interactive analytics dashboard.

Module 5: Machine Learning

Develop models for:

- Tire degradation prediction
- Lap-time prediction
- Pit-stop recommendation
- Race strategy recommendation

Output:

- AI prediction models.

Module 6: Strategy Engine

The user selects:

- Driver
- Circuit
- Tire compound
- Weather
- Current lap

The system recommends:

- Best pit window
- Tire strategy
- Expected race outcome

Output:

- Intelligent strategy recommendation.

Module 7: What-If Simulator

Allow users to:

- Change pit-stop timing.
- Change tire compound.
- Compare strategies.

Output:

- Simulated race results.

Module 8: Dashboard

Provide:

- Charts
- Tables
- Predictions
- Strategy recommendations
- Performance analysis

Output:

- User-friendly web interface.

4. Project Workflow

Data Collection

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Data Cleaning

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Database Storage

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Feature Engineering

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Machine Learning

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Strategy Engine

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Dashboard

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User Recommendations

5. Machine Learning Models

- Linear Regression (Lap-time prediction)
- Random Forest (Tire performance)
- XGBoost (Strategy recommendation)
- Decision Tree (Pit-stop prediction)
- K-Means (Driver performance clustering)

6. Database Tables

- Drivers
- Teams
- Circuits
- Races
- Lap Times
- Tire Data
- Pit Stops
- Weather
- Predictions

7. API Endpoints

Examples:

- Get Drivers

- Get Race Details
- Get Lap Times
- Get Tire Data
- Predict Strategy
- Simulate Race
- Compare Strategies

8. Dashboard Pages

- Home
- Drivers
- Teams
- Circuits
- Race Analytics
- Strategy Recommendation
- What-If Simulator
- Predictions
- About

9. Expected Outputs

- Race analytics
- Tire degradation prediction
- Pit-stop recommendation
- Driver comparison
- Team comparison
- Strategy comparison
- AI recommendations
- Interactive visualizations

10. Final Deliverables

- Source code
- Trained ML models
- Database
- REST API
- Interactive dashboard
- Project report
- README
- Deployment

Features of F1 PitWall AI

1. User Authentication

- User Registration
- User Login & Logout
- Profile Management

2. Formula One Data Collection

- Fetch Driver Information
- Fetch Team Information
- Fetch Circuit Information
- Fetch Race Results
- Fetch Lap-by-Lap Data
- Fetch Pit Stop Data
- Fetch Tire Information

3. Race Analytics Dashboard

- Driver Performance Analysis
- Team Performance Analysis
- Circuit Analysis
- Lap Time Analysis
- Race Timeline Visualization
- Championship Standings

4. Tire Performance Analysis

- Tire Compound Comparison
- Tire Degradation Analysis
- Tire Life Prediction
- Tire Performance Trends

5. Pit Stop Strategy

- Best Pit Stop Window

- Undercut Analysis
- Overcut Analysis
- Pit Stop Time Analysis
- Simulate Different Pit Stop Timings
- Compare Tire Compounds
- Estimate Race Outcomes

6. AI Prediction

- Lap Time Prediction
- Tire Performance Prediction
- Race Position Prediction
- Strategy Recommendation

7. Strategy Recommendation Engine

- Best Tire Strategy
- Best Pit Stop Timing
- Race Strategy Suggestions
- AI-Based Decision Support

8. What-If Race Simulator

- Compare Multiple Strategies

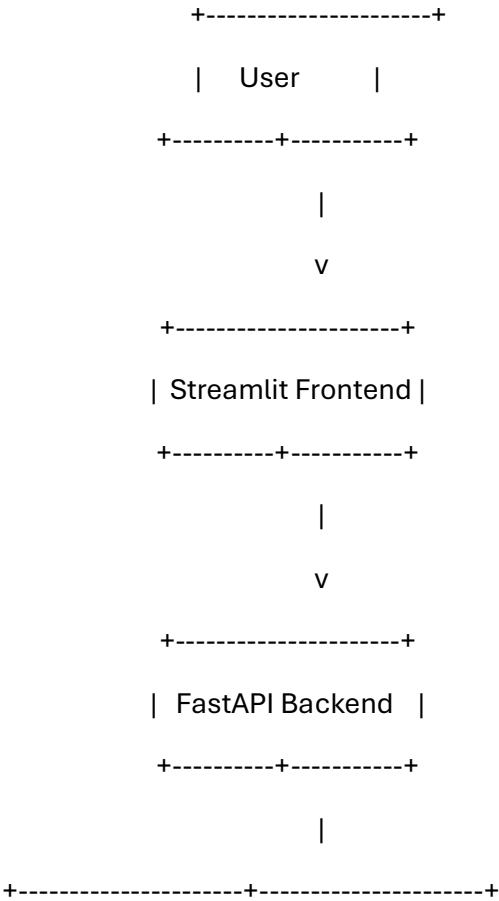
9. AI Race Engineer

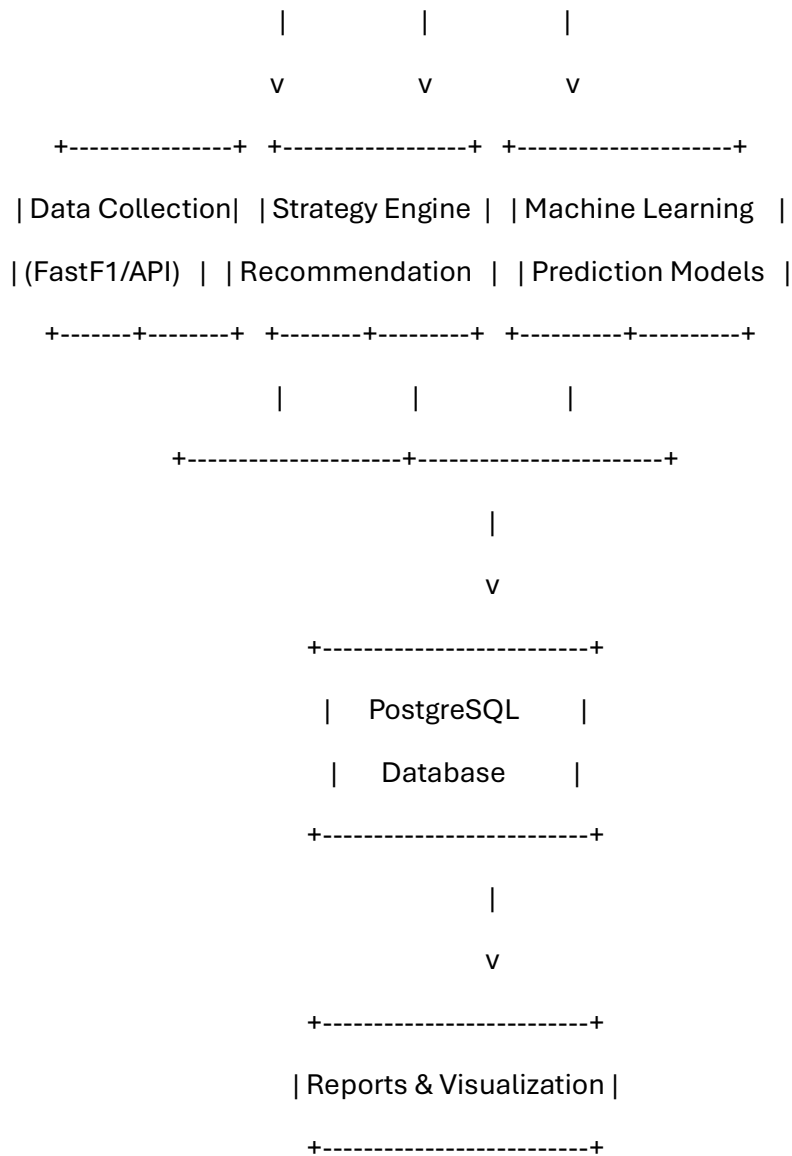
- Ask Strategy Questions
- Explain AI Recommendations
- Driver Performance Insights
- Race Decision Assistance

10. Reports & Visualization

- Interactive Charts
- Race Summary Reports
- Driver Reports
- Team Reports
- Export Reports (PDF/CSV)

System Architecture





Architecture Components

1. Frontend

- Streamlit
- Displays dashboards, charts, predictions, and strategy recommendations.

2. Backend

- FastAPI
- Handles API requests, business logic, and communication with the database and ML models.

3. Data Collection

- Collects historical Formula One data from FastF1 and Ergast API.

4. Database

- Stores race data, driver information, lap times, tire data, and prediction results.

5. Machine Learning Module

- Predicts tire degradation, lap times, and race strategy outcomes.

6. Strategy Engine

- Generates AI-based recommendations for pit stops, tire selection, and race strategy.

7. Reports & Visualization

- Presents interactive charts, analytics, and downloadable reports.

This architecture provides a clear blueprint for how the application is organized and how data flows from collection through analysis to user-facing insights.

Database Design

The **F1 PitWall AI** system uses a relational database to store historical Formula One race data, driver details, team information, race statistics, and AI prediction results. The database is designed to ensure efficient data retrieval, consistency, and scalability.

Database Tables

1. Drivers

Stores information about Formula One drivers.

- Driver ID
- Driver Name
- Driver Number
- Nationality
- Date of Birth
- Team ID

2. Constructors (Teams)

Stores information about Formula One teams.

- Team ID
- Team Name
- Team Principal
- Nationality

3. Circuits

Stores information about Formula One circuits.

- Circuit ID
- Circuit Name
- Location
- Country
- Circuit Length

4. Races

Stores race event details.

- Race ID
- Race Name
- Season
- Round
- Circuit ID
- Race Date

5. Lap Times

Stores lap-by-lap performance data.

- Lap ID
- Race ID
- Driver ID
- Lap Number
- Lap Time
- Sector 1 Time
- Sector 2 Time
- Sector 3 Time

6. Tire Data

Stores tire-related information.

- Tire ID

- Driver ID
- Race ID
- Tire Compound
- Tire Age
- Tire Status

7. Pit Stops

Stores pit-stop details.

- Pit Stop ID
- Driver ID
- Race ID
- Lap Number
- Pit Duration

8. Predictions

Stores AI-generated prediction results.

- Prediction ID
- Driver ID
- Race ID
- Predicted Lap Time
- Predicted Tire Life
- Recommended Pit Lap
- Recommended Strategy

9. Users

Stores registered user information.

- User ID
- Name
- Email
- Password
- Role

Machine Learning Models & AI Algorithms

The **F1 PitWall AI** system uses machine learning algorithms to analyze historical Formula One race data and generate intelligent strategy recommendations. Different algorithms are selected based on the prediction task.

1. Linear Regression

Purpose: Predict lap times based on historical race data.

2. Random Forest

Purpose: Predict tire degradation and estimate tire life.

3. XGBoost

Purpose: Recommend the best race strategy by analyzing multiple race factors such as tire condition, pit-stop timing, weather, and driver performance.

4. Decision Tree

Purpose: Predict the optimal pit-stop window during a race.

5. K-Means Clustering

Purpose: Group drivers based on driving style and race performance.

6. Data Analysis

Purpose: Analyze historical race trends, driver performance, team performance, and circuit statistics to generate meaningful insights.

Input Features

- Driver

- Team
- Circuit
- Lap Number
- Tire Compound
- Tire Age
- Weather Conditions
- Pit Stops
- Lap Times
- Sector Times

Model Outputs

- Predicted Lap Time
- Tire Degradation
- Recommended Pit Stop Window
- Best Tire Strategy
- Driver Performance Analysis
- Race Strategy Recommendation

System Functionalities

The **F1 PitWall AI** system provides the following functionalities:

1. User Management

- User Registration
- User Login and Logout
- User Profile Management

2. Formula One Data Management

- View Drivers
- View Constructors (Teams)
- View Circuits
- View Race Calendar
- View Race Results

3. Race Analytics

- Driver Performance Analysis
- Team Performance Analysis
- Circuit Analysis

- Lap Time Analysis
- Sector Time Analysis
- Championship Standings

4. Tire Performance Analysis

- Analyze Tire Compounds
- Monitor Tire Degradation
- Compare Tire Performance
- Estimate Tire Life

5. Pit Stop Strategy

- Recommend Best Pit Stop Window
- Analyze Pit Stop Performance
- Compare Pit Stop Strategies

6. AI Prediction

- Predict Lap Time

- Predict Tire Performance
- Predict Race Strategy
- Recommend Best Strategy

7. Race Strategy Simulator

- Compare Multiple Race Strategies
- Simulate Different Pit Stop Scenarios
- Analyze Alternative Race Outcomes

8. Dashboard & Reports

- Interactive Charts
- Performance Reports
- Strategy Reports
- Race Summary
- Download Reports

User Roles & Permissions

The **F1 PitWall AI** system consists of two user roles: **Administrator** and **User**.

1. Administrator

The Administrator has full access to the system and is responsible for managing data and monitoring the application.

Permissions

- Login to the Admin Dashboard
- Manage Users
- Manage Driver Data
- Manage Team Data
- Manage Circuit Data
- Update Race Data
- Manage Machine Learning Models
- View System Reports
- Monitor System Performance

2. User

The User can access the analytics and AI features of the application.

Permissions

- Register and Login
- View Driver Information
- View Team Information
- View Circuit Information
- Analyze Race Statistics
- Compare Driver Performance
- View Tire Performance
- View Pit Stop Analysis
- Get AI Strategy Recommendations
- Run "What-If" Race Simulations
- Download Reports
- Update Profile

Project Timeline

Phase	Task	Duration
Phase 1	Requirement Analysis and Project Planning	Week 1
Phase 2	Data Collection from FastF1 and Ergast API	Week 2
Phase 3	Data Cleaning and Preprocessing	Week 3

Phase	Task	Duration
Phase 4	Database Design and Implementation	Week 4
Phase 5	Machine Learning Model Development	Weeks 5–6
Phase 6	Backend Development using FastAPI	Week 7
Phase 7	Frontend Dashboard Development using Streamlit	Week 8
Phase 8	Integration of AI Models and Dashboard	Week 9
Phase 9	Testing and Bug Fixing	Week 10
Phase 10	Deployment and Documentation	Week 11
Phase 11	Final Review and Project Submission	Week 12

The main challenge is that **Formula 1 teams already have highly advanced, proprietary strategy software** developed over many years by engineers, data scientists, and strategists using data that isn't publicly available (live telemetry, tire temperature, fuel load, wind tunnel data, simulator results, etc.). Because of that, it's very unlikely that a team would adopt a student project as-is.

That said, **this project can still open doors** if you position it correctly.

Who you could present it to

Instead of only targeting Formula 1 teams, consider:

- Formula One (F1) as an organization
- F1 teams (Ferrari, Mercedes, McLaren, Red Bull Racing, Williams, Aston Martin, Alpine, Haas, Racing Bulls, Audi)
- Motorsport analytics companies
- Motorsport media organizations
- Universities doing motorsport research
- Recruiters looking for AI, data science, or software engineering talent

What would make them notice?

Your project should demonstrate something valuable, such as:

- A unique strategy optimization approach
- Explainable AI (why the model recommends a strategy)
- High-quality visualizations
- A polished, responsive web application
- Clean, well-documented code
- Strong technical writing and evaluation

Simply recreating race statistics won't be enough.

Build a Professional Portfolio

Treat it like a real product.

Include:

- A professional website
- GitHub repository
- Live demo
- Documentation
- Technical architecture
- Demo video (3–5 minutes)
- Research paper or technical report

Share It Publicly

Publish your work on:

- GitHub
- LinkedIn
- Kaggle
- Medium or Dev.to
- YouTube (demo)
- AI and data science communities

A well-presented project often attracts more attention than one that's never shared.

Reach Out Professionally

If the project is mature, you can contact engineering or data teams with a concise message that includes:

- A brief description
- GitHub link
- Live demo
- Technical report
- What makes your approach different

Expect that many organizations won't respond, but a thoughtful, polished project gives you a much better chance of getting noticed.

Could an F1 team actually use it?

Probably **not directly**, because they rely on private data and highly specialized internal systems.

However, your project could still demonstrate that you understand:

- Data engineering
- Predictive modeling
- Decision support systems
- Software architecture
- Sports analytics

Those are valuable skills across many industries.